

Lab - Running a Simple Container (Docker)

To Begin with the Lab

Summary

This lab demonstrates how to run a Docker container from an existing Docker image. It explains the role of Docker Desktop, Docker Hub, Docker images, containers, and how to use Docker commands to download and run an Nginx web server inside a container.

Understanding

Docker Desktop provides a graphical interface (GUI) and installs the Docker Engine (Docker runtime) on your computer. The Docker Engine is responsible for downloading Docker images, creating containers, and managing them.

A **Docker Image** is a read-only template that contains everything needed to run an application. A **Docker Container** is a running instance of that image.

In this lab, the **Nginx** image is downloaded from **Docker Hub**, and Docker creates a container from that image. The container runs the Nginx web server without installing Nginx directly on the operating system.

Port mapping allows users to access applications running inside the container from the host machine. For example, mapping **Host Port 8080** to **Container Port 80** makes the Nginx website available at `http://localhost:8080`.

Example

- Docker Image: `nginx:latest`
 - Docker Container: Running instance of the Nginx image
 - Host Port: 8080
 - Container Port: 80
 - Access URL: `http://localhost:8080`
-

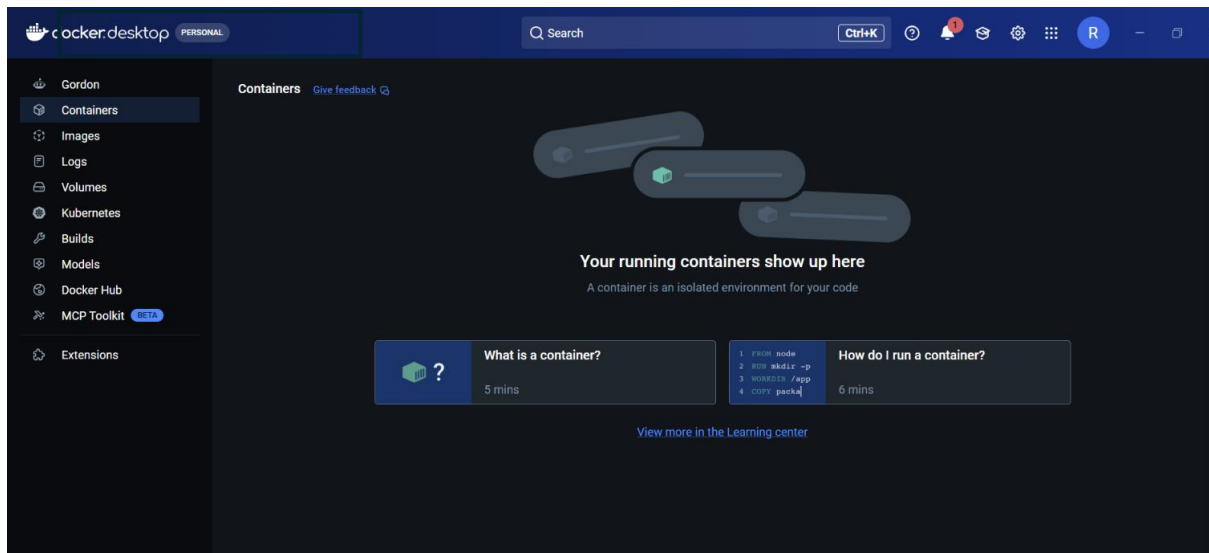
Lab Steps

Step 1: Open Docker Desktop

- Launch Docker Desktop.
- Wait until Docker Engine is running.
- Ensure Docker status shows **Running**.

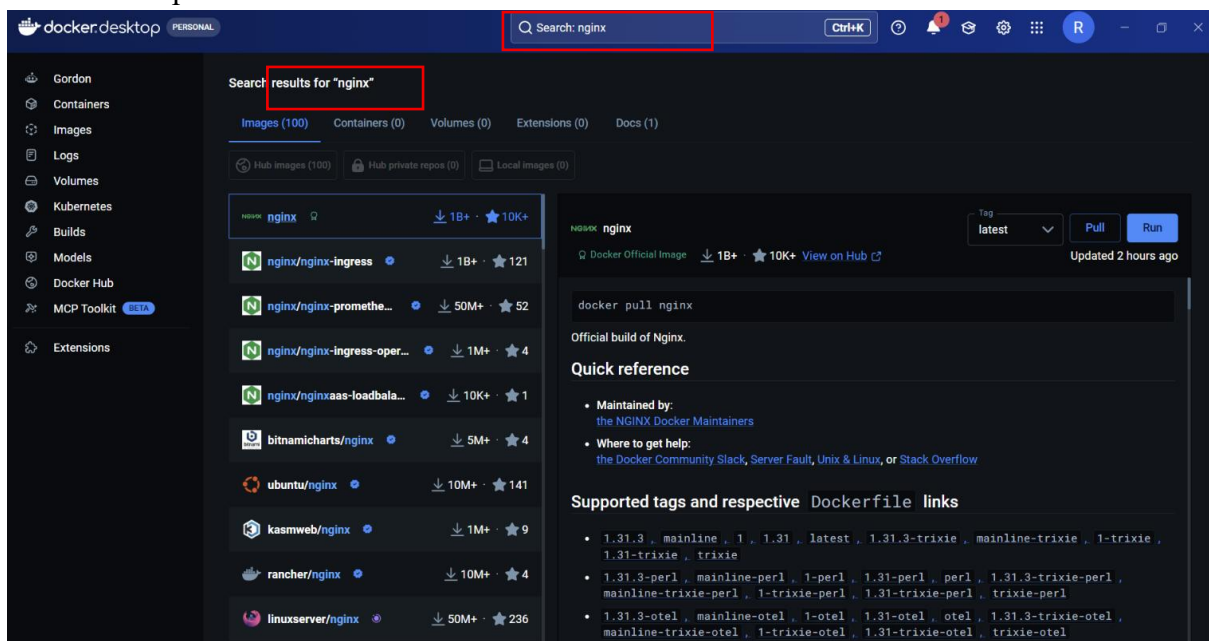
Step 2: Create a Docker Hub Account

- Open <https://hub.docker.com>.
- Sign up using Email or Google account.
- Verify your email.
- Sign in to Docker Hub.



Step 3: Search for the Nginx Image

- Search for **nginx**.
- Open the **Official Nginx Image**.
- Review available image tags.
- Note the pull command.



Step 4: Open Command Prompt

- Open Command Prompt or PowerShell.
- Verify Docker is installed.

```
docker --version
```

```
Windows PowerShell
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PS C:\Users\rrath> docker --version
Docker version 29.6.1, build 8900f1d
PS C:\Users\rrath> |
```

Step 5: Run an Nginx Container

Execute the following command:

```
docker run -d --name nginx-container -p 8080:80 nginx
```

Command Explanation

- `docker run` → Creates and starts a container.
- `-d` → Runs the container in background (Detached mode).
- `--name nginx-container` → Assigns a container name.
- `-p 8080:80` → Maps Host Port 8080 to Container Port 80.
- `nginx` → Docker image name.

```
Windows PowerShell
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PS C:\Users\rrath> docker --version
Docker version 29.6.1, build 8900f1d
PS C:\Users\rrath> docker run -d --name nginx-container -p 8080:80 nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
d26f27cc8c41: Pull complete
2bedaf25031a: Pull complete
b6698f04e005: Pull complete
062e450697fa: Pull complete
3c7ab7949321: Pull complete
82454cddb456: Pull complete
cacfcdd01f30: Pull complete
6c496f5b5050: Download complete
eald76ccc2c6: Download complete
Digest: sha256:5a88c9c45479443d7be2eadc894b4ed0a9801bae03d97a5760ae13b5c2005942
Status: Downloaded newer image for nginx:latest
7fc8d854ef9c6df150c2b112d32213870440efc866225fdd6a7dc0fc9d582a33
PS C:\Users\rrath> |
```

Step 6: Verify Running Containers

List running containers.

Execute the following command:

```
docker ps -a
```

Expected output includes:

- Container ID
- Image Name

- Container Name
- Status
- Port Mapping

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\rrath> docker --version
Docker version 29.6.1, build 8900f1d
PS C:\Users\rrath> docker run -d --name nginx-container -p 8080:80 nginx
Unable to find image 'nginx:latest' locally
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2bedaf25031a: Pull complete
b6698f04e005: Pull complete
062e450697fa: Pull complete
3c7ab7949321: Pull complete
82454cdbf456: Pull complete
cacfcdd01f30: Pull complete
6c496f5b5050: Download complete
eald76ccc2c6: Download complete
Digest: sha256:5a88c9c45479443d7be2eadc894b4ed0a9801bae03d97a5760ae13b5c2005942
Status: Downloaded newer image for nginx:latest
7fc8d854ef9c6df150c2b112d32213870440efc866225fdd6a7dc0fc9d582a33
PS C:\Users\rrath> docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS
7fc8d854ef9c   nginx    "/docker-entrypoint..." 4 minutes ago  Up 4 minutes  0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
PS C:\Users\rrath>
```

Step 7: View Downloaded Images

Display all images stored locally.

Execute the following command:

docker images

This shows:

- Repository
- Tag
- Image ID
- Size

```
Windows PowerShell
PS C:\Users\rrath> docker images
IMAGE          ID              DISK USAGE  CONTENT SIZE  EXTRA
nginx:latest    5a88c9c45479    241MB       66MB          U
PS C:\Users\rrath>
```

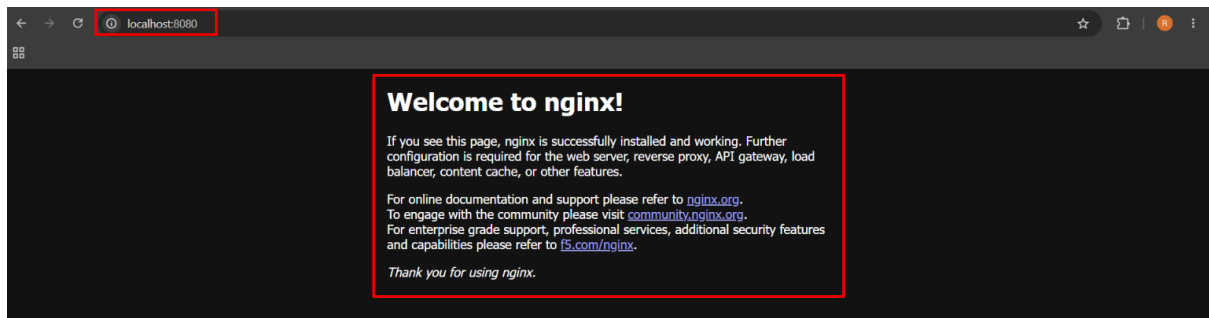
Step 8: Access the Nginx Website

Open a web browser.

Visit:

http://localhost:8080

The default **Welcome to nginx!** page should appear.

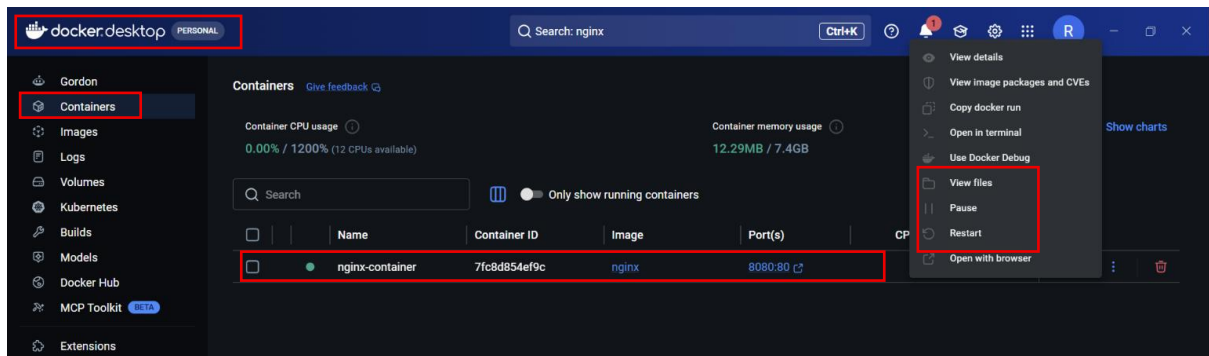


Step 9: Manage Containers Using Docker Desktop

Open Docker Desktop.

Under **Containers**, you can:

- View running containers.
- Stop a container.
- Restart a container.
- Pause a container.
- Delete a container.



Step 10: View Docker Images

Go to the **Images** section in Docker Desktop.

Verify that the **nginx** image is available locally.

